

The intestinal microbiome: a separate organ inside the body with the metabolic potential to influence the bioactivity of botanicals.

For many years, it was believed that the main function of the large intestine was the resorption of water and salt and the facilitated disposal of waste materials. However, this task definition was far from complete, as it did not consider the activity of the microbial content of the large intestine. Nowadays it is clear that the complex microbial ecosystem in our intestines should be considered as a separate organ within the body, with a metabolic capacity which exceeds the liver with a factor 100. The intestinal microbiome is therefore closely involved in the first-pass metabolism of dietary compounds. This is especially true for botanical supplements, which are now marketed for various health applications. Being of natural origin, their structural building blocks, such as polyphenols, are often highly recognized by the human and especially the intestinal microbial metabolism machinery. Intensive metabolism results in often low circulating levels of the original products, with the consequence that final health effects of botanicals are often related to specific active metabolites which are produced in the body rather than being related to the product's original composition. Understanding how such metabolic processes contribute to the in situ exposure is therefore crucial for the proper interpretation of biological responses. A multidisciplinary approach, characterizing the food and phytochemical intake as well as the metabolic potency of the gut microbiota, while measuring biomarkers of both exposure and response in target tissues, is therefore of critical importance. With polyphenol metabolism as example, this review describes how the incorporation of microbial metabolism as an important variable in the evaluation of the final bioactivity of botanicals strongly increases the relevance and predictive value of the outcome. Moreover, knowledge about intestinal processes may offer innovative strategies for targeted product development. Copyright © 2010 Elsevier B.V. All rights reserved.